Experiment:3

MATLAB Simulation of Superheterodyne Receiver for AM Communication System with Frequency Mixing and Filtering.

Aim: To simulate a superheterodyne receiver for an AM communication system using MATLAB. To study frequency mixing, intermediate-frequency (IF) filtering, and recovery of the original message signal.

Software Required:MATLAB

Program:

```
clc; clear; close all;

Fs = 200000;
t = 0:1/Fs:0.005;
Ac = 10; Am = 5;
fc = 20000; fm = 1000; fIF = 5000;
m = Am/Ac;
fLO = fc-fIF;

msg = Am*cos(2*pi*fm*t);
AM = Ac*(1+m*cos(2*pi*fm*t)).*cos(2*pi*fc*t);
mix = AM.*cos(2*pi*fLO*t);

N = length(t);
f = (-N/2:N/2-1)*Fs/N;

X = fftshift(fft(mix));
X(abs(f)<fIF-1500 | abs(f)>fIF+1500) = 0;
IF = real(ifft(ifftshift(X)));

Y = fftshift(fft(IF.*cos(2*pi*fIF*t)));
Y(abs(f)>2000) = 0;
out = real(ifft(ifftshift(Y)));
rec = (out-mean(out))*4/(Ac*m);

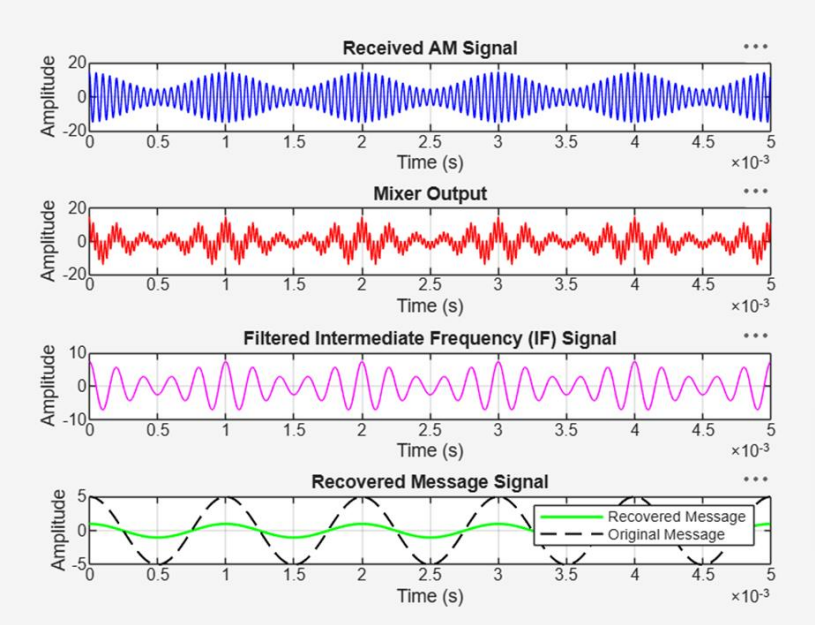
subplot(4,1,1);
plot(t,AM,'b'); grid on;
title('Received AM Signal'); ylabel('Amplitude');

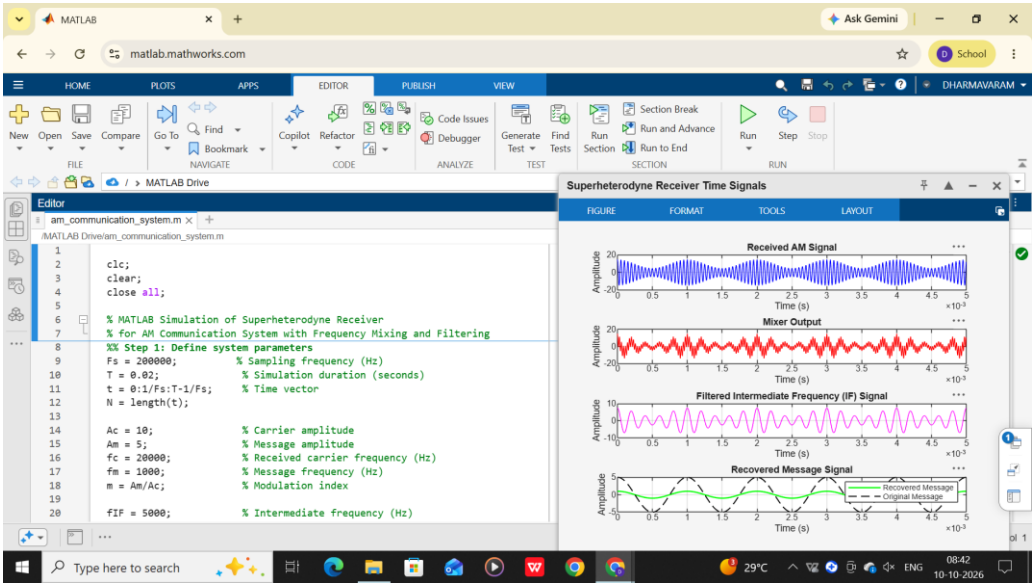
subplot(4,1,2);
plot(t,mix,'r'); grid on;
title('Mixer Output'); ylabel('Amplitude');

subplot(4,1,3);
plot(t,IF,'m'); grid on;
title('Filtered Intermediate Frequency (IF) Signal');
ylabel('Amplitude');
```

```
subplot(4,1,4);
plot(t,rec,'g','LineWidth',1.2); hold on;
plot(t,msg,'k--','LineWidth',1.2);
grid on;
title('Recovered Message Signal');
xlabel('Time (s)'); ylabel('Amplitude');
legend('Recovered Message','Original Message')
```

Output:





Result: The superheterodyne AM receiver was successfully simulated using MATLAB. Frequency conversion, IF filtering, and recovery of the original message signal were demonstrated.